CAPSTONE PROJECT

Predicting Online Shoppers' Purchase Intention

A k-Nearest Neighbors Classifier

Dataset: UCI Online Shoppers Purchasing Intention • 12,330 sessions • 17 features

May 2026

How This Talk Is Organized

From the question to the answer to the caveats.

1	**Problem**	What are we trying to predict?
2	**Data**	What does the dataset look like?
3	**EDA**	What do the features tell us?
4	**Method**	How does kNN work, and how did we build it?
5	**Tuning**	How did we choose the value of k?
6	**Results**	How well does it actually perform?
7	**Validation**	Cross-validation and baseline comparison.
8	**Limits**	Where the model breaks down.
9	**Conclusion**	What we learned and what comes next.

The Problem

Most online sessions never become a purchase. Can we predict which ones will?

Business Question

- E-commerce sites get millions of sessions, but only a small fraction convert.
- Knowing in advance which sessions are likely to buy lets the business intervene — discounts, live chat, retargeting.
- Goal: given a session's behavior and metadata, predict whether it will end in a purchase.

ML Framing

- Binary classification: 1 = purchase, 0 = no purchase.
- Highly imbalanced — only 15.47% of sessions are positives.
- Therefore: accuracy alone is misleading. Must report precision, recall, and F1.
- Algorithm of choice: k-Nearest Neighbors, implemented from scratch in NumPy.

The Dataset

UCI Online Shoppers Purchasing Intention — 12,330 real e-commerce sessions.

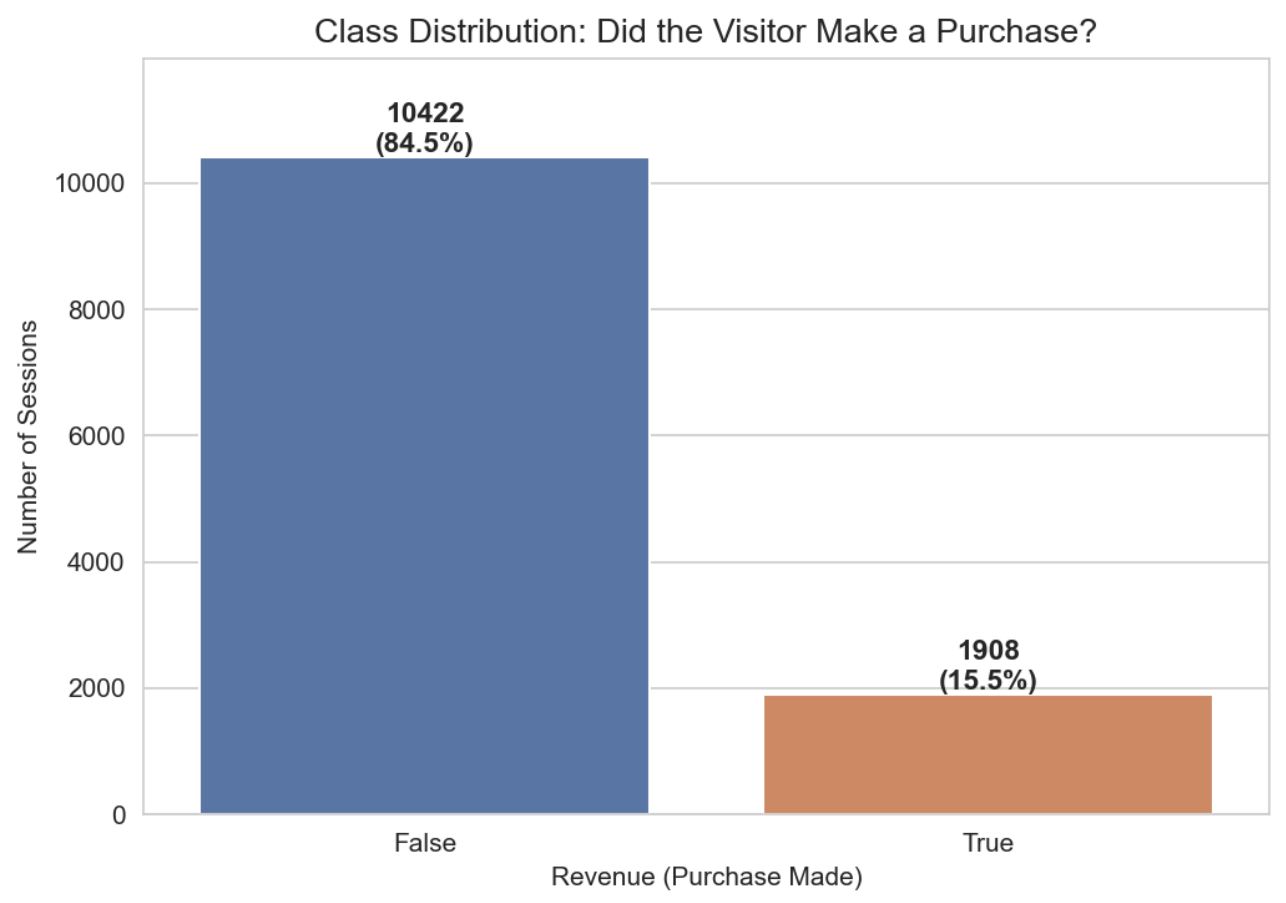
12,330	17	15.47%	5.46 : 1
Sessions in dataset	Predictor features	Sessions that purchase	No-buy : Buy ratio

Feature categories

Page visits	Administrative, Informational, ProductRelated	*How many pages of each kind were viewed.*
Time spent	*_Duration columns	*Total seconds on each page type.*
Google Analytics	BounceRates, ExitRates, PageValues	*Site-side metrics; PageValues is highly predictive.*
Calendar	SpecialDay, Month	*Closeness to holidays; visit month.*
Tech metadata	OperatingSystems, Browser, Region, TrafficType	*User connection profile.*
Visitor info	VisitorType, Weekend	*Returning vs. new; weekend flag.*

For Every 1 Buyer, ~5.5 People Just Browsed and Left

This is normal for e-commerce, but it shapes the entire evaluation strategy.

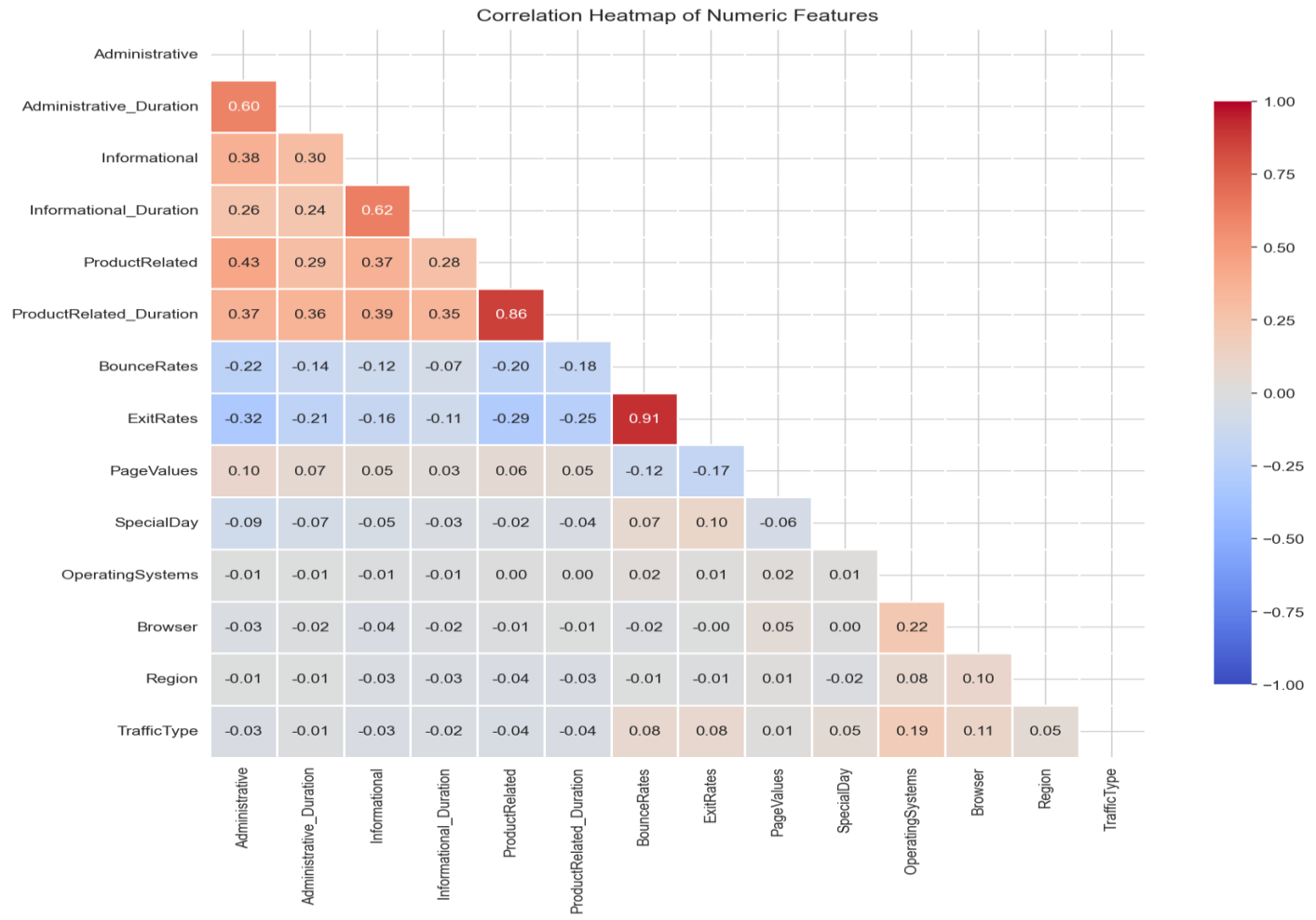


The "Accuracy Trap"

- A model that always says "No-Buy" is right 84.5% of the time.
- That’s an 84% accuracy with zero learning.
- So accuracy alone is not a passing grade.
- We commit upfront to reporting precision, recall, F1, and the confusion matrix.

Some Features Are Saying the Same Thing

Multicollinearity matters for kNN — duplicate signals get double-counted in distance.



Notable Pairs

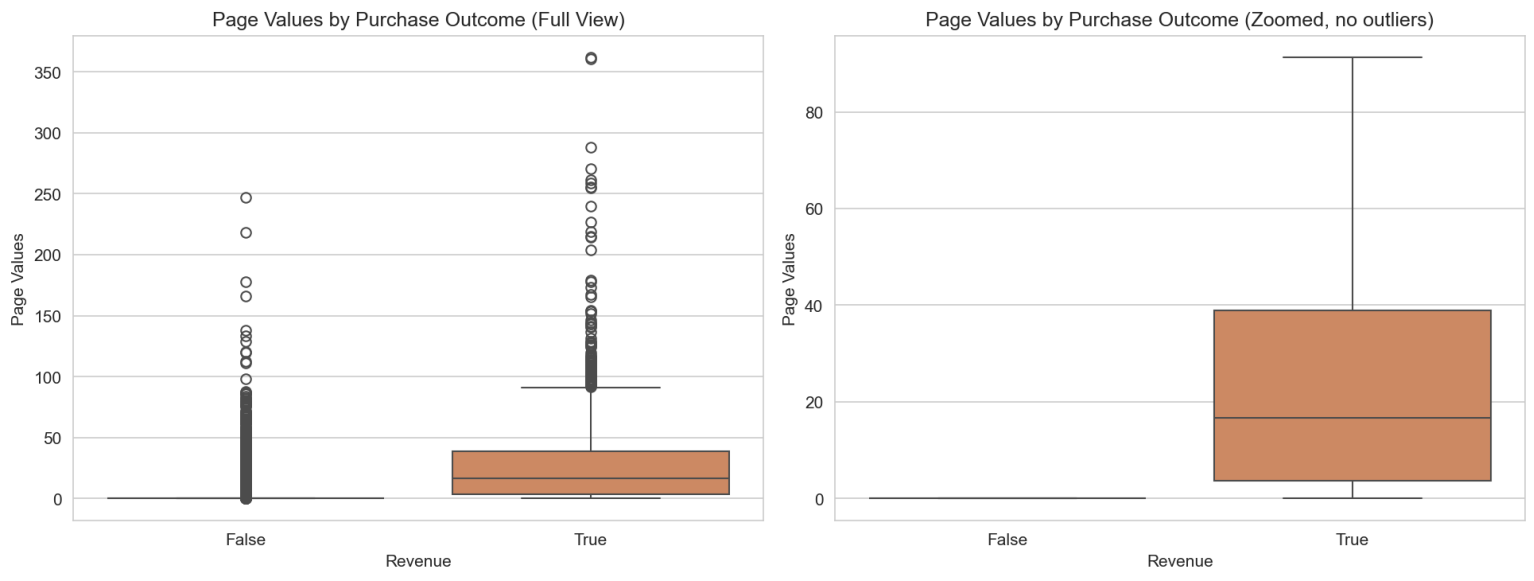
- BounceRates ↔ ExitRates: r = 0.913
- ProductRelated ↔ ProductRelated_Duration: r = 0.861

Why It Matters

- kNN measures distance using all features at once.
- Duplicate features inflate that behavior's influence on neighbor selection.

PageValues: The Single Most Predictive Feature

Buyers arrive at high-value pages; non-buyers don't.



The Numbers

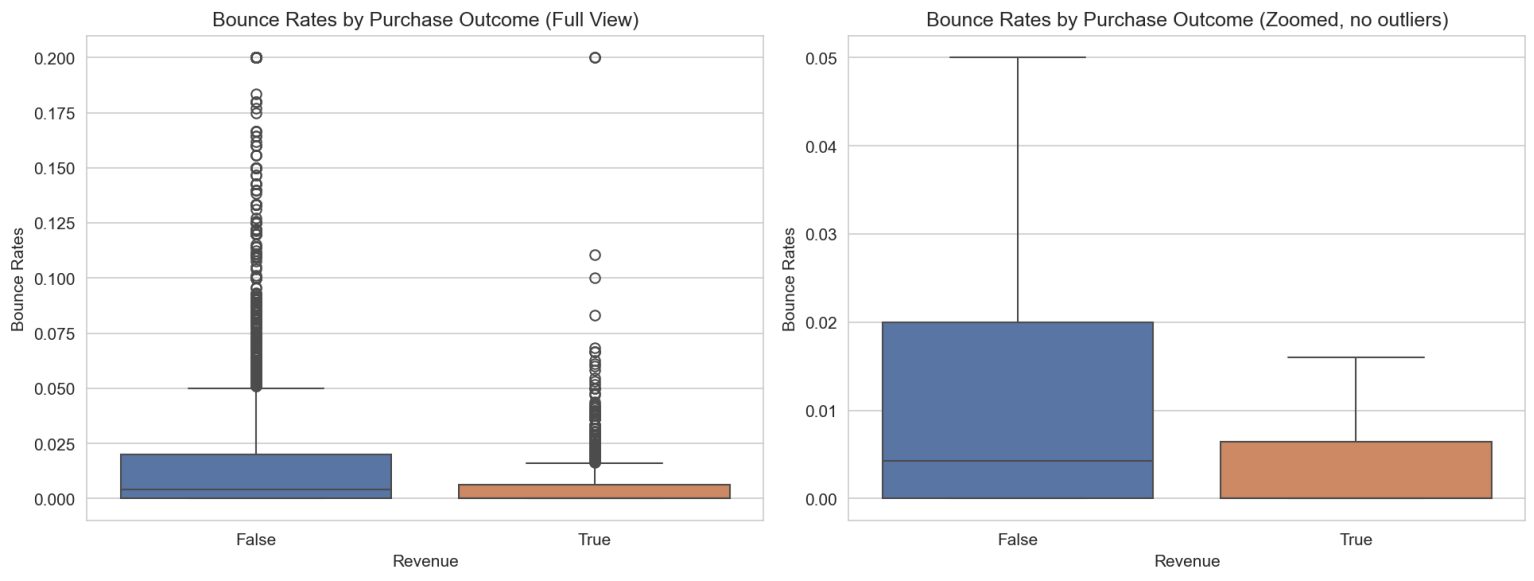
- Buyer mean: 27.26
- Non-buyer mean: 1.98
- 13.8× difference
- Non-buyer median: literally 0

Why It Matters

- Strong class separability is exactly what kNN exploits.

Buyers Stay. Non-Buyers Drift Away.

BounceRate captures the difference between intent and casual browsing.



Engagement Gap

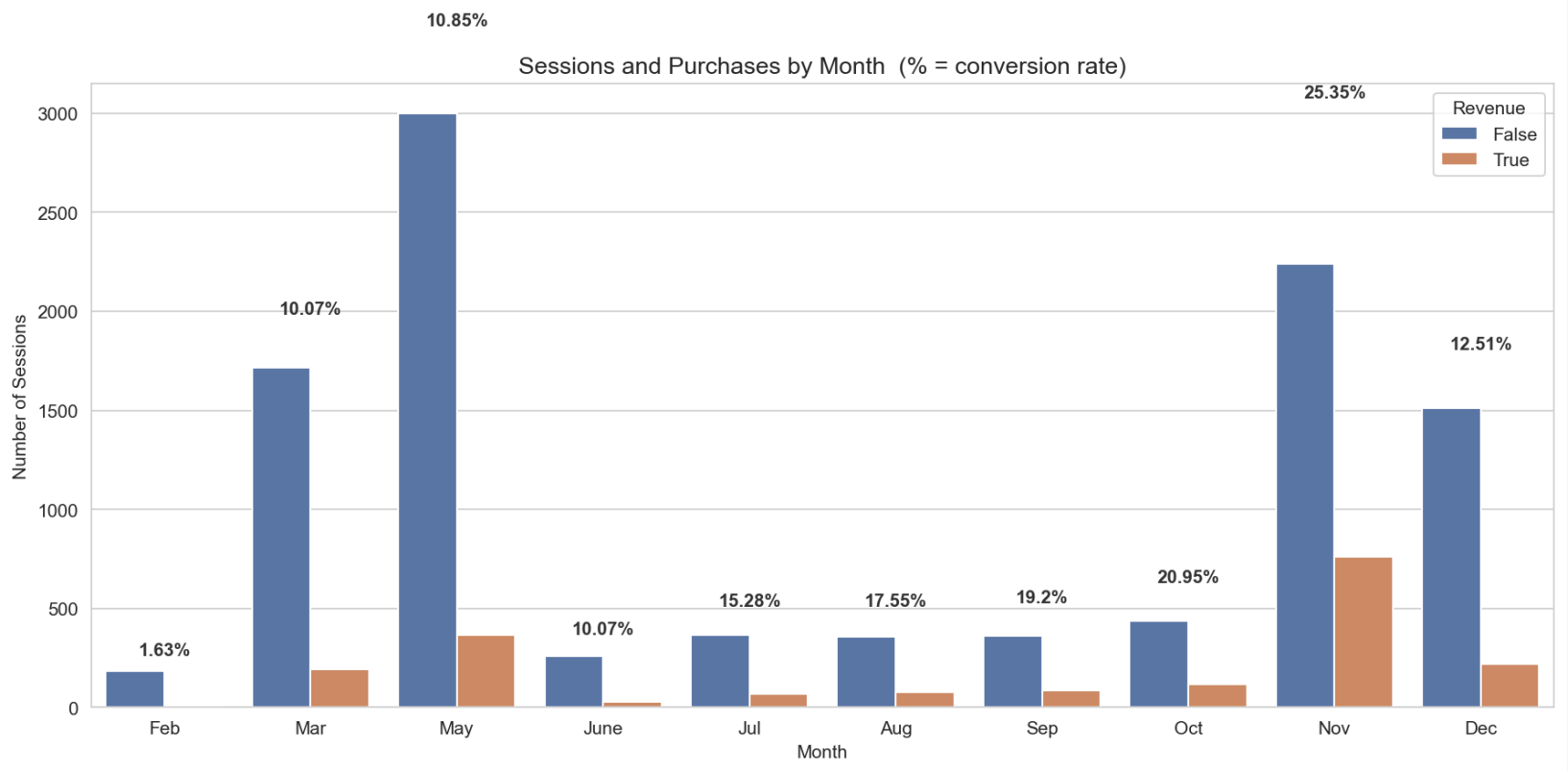
- Buyer mean bounce: 0.0051
- Non-buyer mean: 0.0253
- Roughly 5× higher for non-buyers
- Buyer median: 0

Implication

- Engagement depth is strongly tied to purchase completion.

November Is the Month That Matters

Conversion rate isn't flat across the year. It spikes during holiday shopping.



Conversion by Month

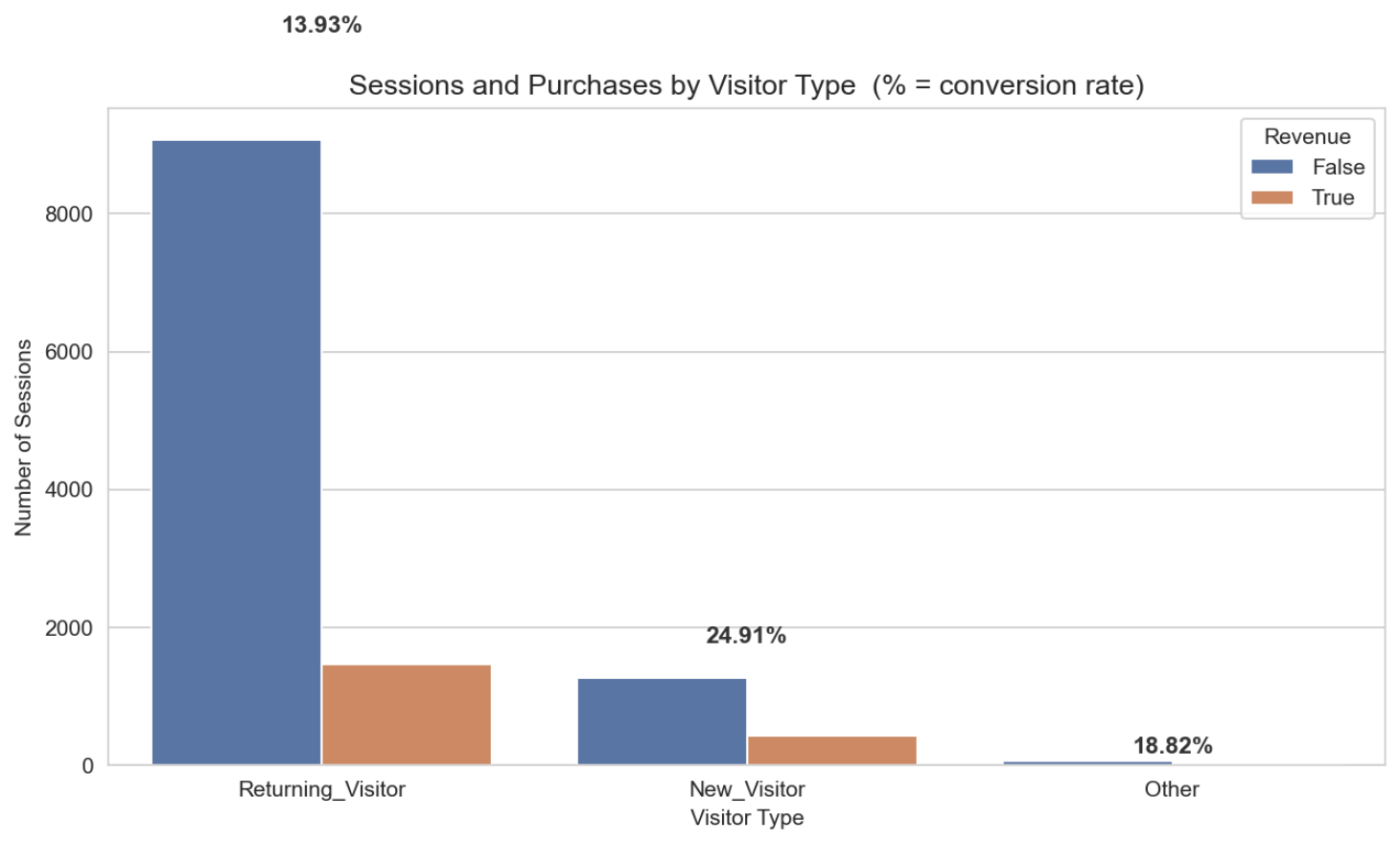
- Feb: 1.63%
- Mar: 10.03%
- May: 10.85%
- Jul: 15.28%
- Sep: 19.07%
- Nov: 25.35% ← peak

Takeaway

- Month one-hots give the model a strong prior.

New Visitors Convert Better Than Returning Visitors

A surprising result — and exactly the kind of insight a stakeholder would ask about.



Conversion Rate

- New visitors: 24.91%
- Returning visitors: 13.93%
- Almost 2× higher for new visitors

Why?

- New visitors arrive with high intent (clicked an ad, searched for the product).
- Returning visitors include a lot of casual browsers who keep coming back.

k-Nearest Neighbors in One Picture

No training step. Just measure distance to every known example and vote.

1 Standardize

Subtract the training mean and divide by the training std for every feature, so no feature dominates by sheer scale.

2 Measure distance

For a new session, compute Euclidean distance to every training point: $\sqrt{\Sigma(x_i - t_i)^2}$.

3 Pick the k closest

Sort by distance and keep the k smallest — those are the k most similar past sessions.

4 Majority vote

Look at their Revenue labels. Whichever class appears more often wins. That’s the prediction.

The Pipeline We Built — Seven Scripts Estimators

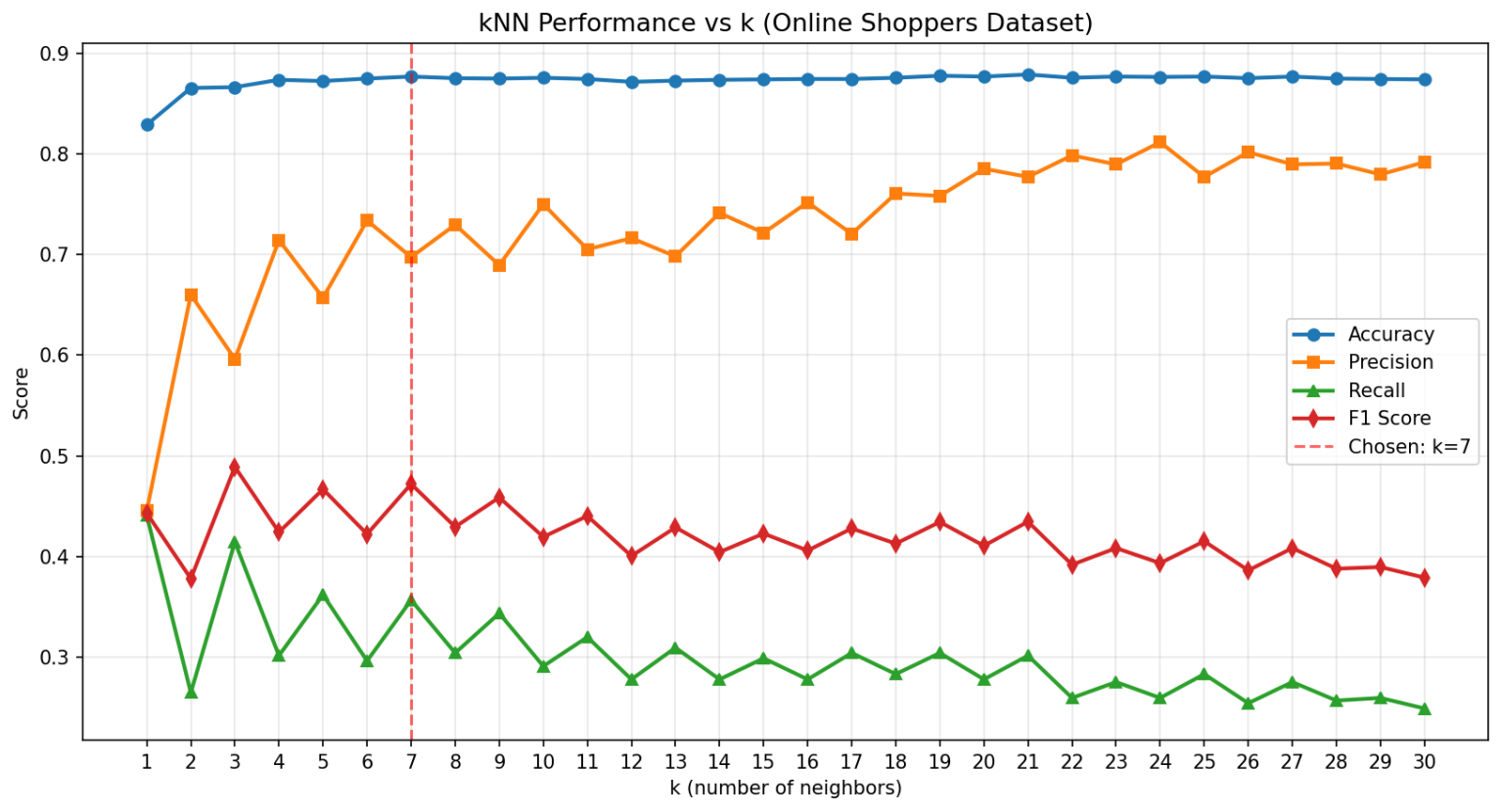
Every step was implemented from NumPy to make the algorithm visible.

Key engineering choices

- Stratified 80 / 20 split — preserves the 84.5 / 15.5 class ratio in both partitions.
- Standardization fitted on train ONLY — no test-set information leaks into the scaler.
- Vectorized Euclidean distance with NumPy broadcasting — fast even at 9,865 × 28.
- Manual confusion matrix and metrics — no sklearn.metrics shortcut.

Sweeping k from 1 to 30

F1 peaks at k = 3; we chose k = 7 for its higher precision. Recall and precision pull in opposite directions.



Best k by metric

- Best Accuracy: k = 21
- Best Precision: k = 24
- Best Recall: k = 1
- Best F1: k = 3

Chosen: k = 7 (higher precision)

How we chose k

- F1 to find the peak region (k ≈ 3–9), avoiding both extremes.
- Precision to pick within it → k = 7 — a confident, conservative classifier.

Final Test-Set Performance (k = 7)

Held-out 20% test set, 2,465 sessions, never seen during tuning preparation.

87.67%	0.6974	0.3570	0.4722
Accuracy	Precision	Recall	F1 Score

How to read these numbers

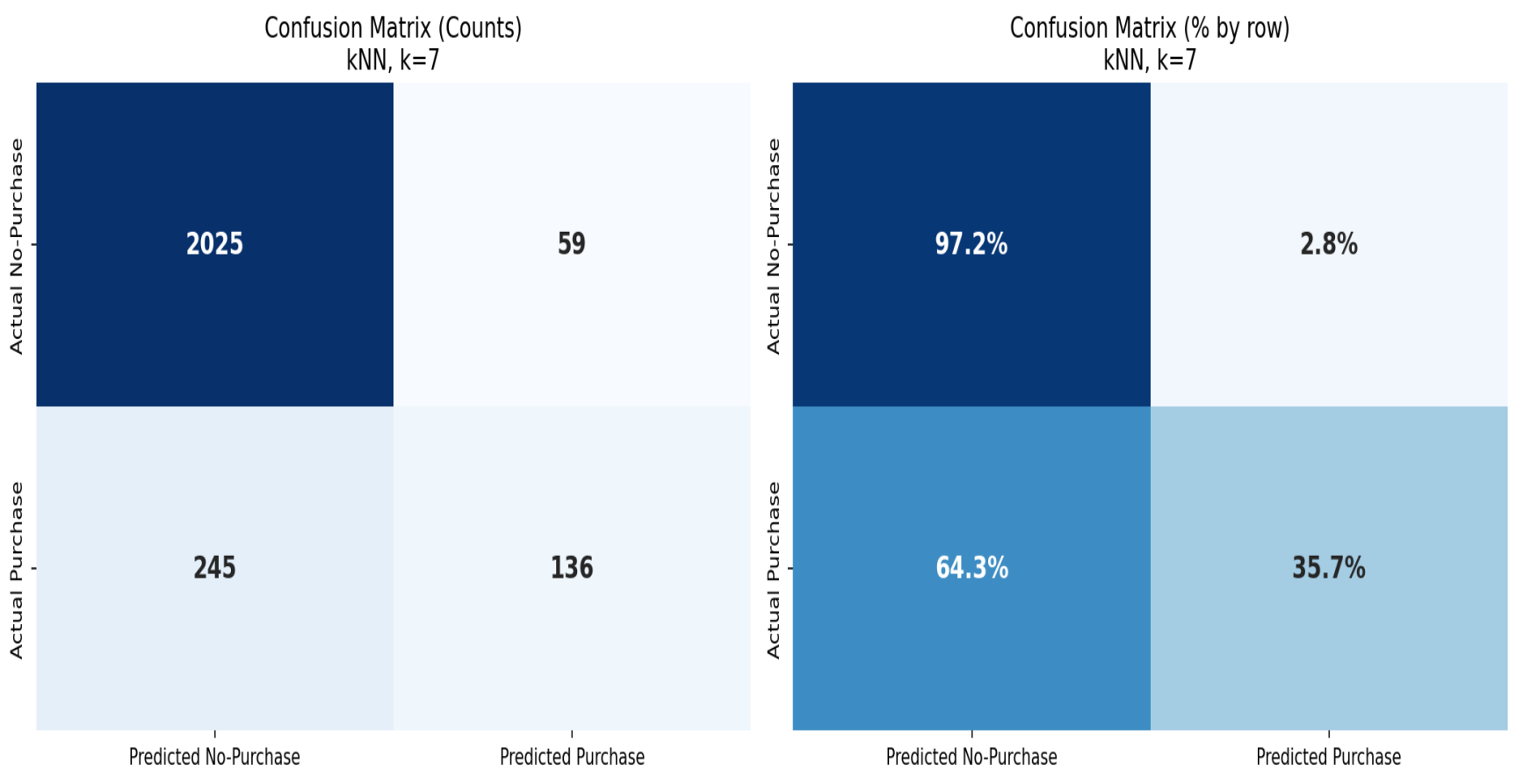
- Precision 0.70 → when we say "buyer", we're right 70% of the time.
- Recall 0.36 → we catch only 36% of the actual buyers.
- High precision, lower recall — a conservative classifier.

Confusion matrix counts

- TP = 136 (correctly flagged buyers)
- FP = 59 (predicted buy, didn't)
- FN = 245 (missed buyers — recall pain)
- TN = 2,025 (correctly excluded non-buyers)

Confusion Matrix — Counts and Row Percentages

Where the model is right, and where it pays the cost of class imbalance.



In Plain English

- Most non-buyers are correctly identified (97.2%).
- Most buyers slip through (we catch only 35.7%).
- When we do flag a buyer, we're usually right.

Validation: Stable Across Folds — but a Simple Rule Beats Us

5-fold stratified CV confirms stability; but a one-feature PageValues rule outperforms our kNN.

5-fold Stratified Cross-Validation

- Stratified folds preserve the 84.5 / 15.5 class ratio in every fold.
- Each fold gets its own train-only standardization (no leakage).
- Per-fold metrics are tightly clustered.
- Conclusion: the test-set numbers are not a lucky split.

Baseline Comparison

Model	Accuracy	F1
Always "No-Purchase"	0.8454	0.0000
Rule on PageValues	0.8929	0.6819
kNN (k = 7) — our model	**0.8767**	**0.4722**

The PageValues rule beats kNN on F1 — kNN dilutes the dominant signal across 28 features. A humbling, instructive result.

What the Numbers Mean for the Business

A high-precision, lower-recall classifier — useful for some interventions, weak for others.

Good Fit If…

- Acting on a non-buyer is expensive (e.g., heavy discount, live agent).
- You want fewer-but-confident alerts, not blanket coverage.
- You can pair this model with low-cost site-wide nudges that catch the rest.

Bad Fit If…

- Missed buyers are far more costly than wasted offers.
- The product needs blanket personalization (recall matters more than precision).
- You need calibrated probabilities, not hard predictions.

Honest Limitations of This Model

A capstone is judged by candor as much as by score.

Imbalance not treated	No SMOTE, no class weighting, no threshold tuning, no distance-weighted voting. The recall ceiling is largely a consequence of this.
Multicollinearity left in	BounceRates / ExitRates (r = 0.913) and ProductRelated / *_Duration (r = 0.861) double-count behavior in distance.
One-hot inflates the distance space	13 binary one-hot columns sit alongside 14 continuous ones in the same Euclidean geometry — distorts neighbor selection slightly.
A one-feature PageValues rule beats the full model	kNN doesn't do feature selection.
Only Euclidean + uniform vote	No Manhattan / Mahalanobis / weighted-vote experiments. Distance-weighted voting in particular is known to help on imbalanced data.
No ROC / probability output	Reported only point classification metrics. ROC + AUC would visualize the full precision-recall trade-off.

What We Would Try Next

Concrete improvements that follow directly from the limitations on the previous slide.

1	**Distance-weighted voting**	Closer neighbors count more. Cheap to add; usually the single biggest win on imbalanced data.
2	**Threshold tuning on the kNN score**	Use k_pos / k as a soft probability and pick the threshold by Youden's J or by business cost.
3	**Resampling**	SMOTE oversampling of the minority class (or principled under-sampling of the majority).
4	**Feature redundancy cleanup**	Drop ExitRates (keep BounceRates) and ProductRelated_Duration (keep ProductRelated), or run PCA.
5	**Cleaner tuning protocol**	Hold out a true validation split or use nested CV so the test set is touched exactly once.
6	**Compare against tuned baselines**	Logistic regression, random forest, gradient boosting — to position kNN against the literature on this dataset.

Five Things to Take Away

1. A kNN classifier can learn real, transferable signal from web-session data.
2. On the UCI Online Shoppers data, k = 7 gave 87.67% accuracy and 0.47 F1 on a held-out test set.
3. Class imbalance is the single most important factor shaping these numbers.
4. High precision (0.70), low recall (0.36) — choose the algorithm that matches your cost structure.
5. The biggest improvements aren’t in tuning more k values. They are in handling imbalance, removing redundant features, and using a clean validation protocol.

Thank You

Questions & Discussion

Predicting Online Shoppers' Purchase Intention • kNN